

17.11.16 (E).

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
Semester: Nov 2016

Max. Marks: 100

Duration: 3 Hrs

End Semester Exam

Class: **SY.BTech**
Name of the Course: OOPM
Course Code: UCEC302

Semester: III
Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Make suitable assumptions if required and state them clearly
- (3) All programs and program segments should be strictly in JAVA

Question No.		Max. Marks
Q1(a)	Draw class diagram for withdrawing money from the ATM.	[05]
Q1(b)	Write short notes on super keyword.	[05]
Q1(c)	Write short note on cohesion and coupling.	[05]
Q1(d)	Show the use of toString() method with example.	[05]
Q2(a)	Write a program to display area of square and rectangle. Make use of interface to define templates of methods to be there in the derive classes. OR What is an abstract class? How it is different from interface explain with an example.	[10]
Q2(b)	What are the benefits of packages? What are the Java API packages? Write a program to demonstrate the creation and use of a package.	[10]
Q3(a)	What is an exception? How exceptions are handled in JAVA, explain by giving examples. OR Write a program in java to accept the values of a, b, c, d and calculate and display $((a*b) + (b*c)) / (b*d)$. Create user defined exception to display proper message when value of $(b*d)$ is ZERO.	[10]

Q3(b)	Explain method overloading and method overriding with suitable example.	[10]
Q4 (a)	With the help of suitable example explain multithreading in terms of following:- (i) Creating threads, extending the thread class. (ii) Life-cycle of Thread. OR Write a program to display the following using two threads. \$1\$2\$3\$4\$5\$6\$7\$8\$9	[10]
Q4 (b)	What is an applet? Explain life cycle of an applet and write an applet program to draw a smiling face.	[10]
Q5 (a)	Write a program in JAVA that computes the sum of a list of integers that is supplied by a user. The end of data signaled by the value -999. This value is used only as a flag and not used in sum. OR Write a program to accept a number from command line and to check whether it is Armstrong Number or Not.	[10]
Q5 (b)	Elaborate on the usage of final keyword, finally and finalize() method with example.	[10]